Twenty-seven maximum independent sets in triangular grid graphs

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Abstract

Let $G_3(d)$ have as vertices the nonnegative integer triples of sum d, with two vertices adjacent when their ℓ_1 distance is 2. We prove that $G_3(d)$ has exactly 27 maximum independent sets whenever $d \geq 10$ and $3 \nmid d$. We also describe all of them: each agrees with one of the three natural color classes away from the three radius-three corners, and each color class has exactly nine corner extensions. The proof uses an explicit rational triangle cover with 89 weights. A coordinate reduction verifies the cover for every allowed d; saturation and a balancing argument then reduce the classification to three ten-vertex problems. Together with Machacek's uniqueness theorem for $3 \mid d$, the result gives Howroyd's eventual period-three formula.

1 Introduction and statement

For $d \geq 0$, put

$$V_d = \{(x_1, x_2, x_3) \in \mathbb{Z}^3_{\geq 0} : x_1 + x_2 + x_3 = d\}.$$

The graph $G_3(d)$ has vertex set V_d, with distinct vertices u, v adjacent if and only if $\|u - v\|_1 = 2$. Write $\alpha_3(d)$ for its independence number and $a_3(d)$ for the number of independent sets of that size. Sets are counted with their coordinates: no quotient by rotations, reflections, or graph isomorphisms is taken. This is also the monomial graph used by Machacek [2], since, for $u, v \in V_d$,

$$\deg \operatorname{lcm}(x^u, x^v) = d + \tfrac{1}{2}\|u - v\|_1.$$

Howroyd's conjecture, recorded in OEIS A297557 on 1 January 2018 [1]

and stated as Conjecture 1.1 in [2], is

$$a_3(d) = \begin{cases} 1, & 3 \mid d, \\ 27, & 3 \nmid d \end{cases} \qquad (d \geq 9). \tag{1}$$

Machacek's Theorem 2.1 establishes the divisible case. Appendix A records a local correction to a deletion statement in the cited arXiv version; it is separate from our proof of the nondivisible case.

The independence number is already known. Machacek [2, Section 2] attributes its formula to Geramita, Gregory, and Roberts. For $d \geq 10$ with $3 \nmid d$, it is $\alpha_3(d) = \binom{d+2}{2}/3$. The same size result appears in Zabokritskiy's Theorem 4.2 [3] and in Proposition 17 of Kreindel, Barouch Essayag, and Zabokritskiy [4]. Corollary 18 of the latter paper gives uniqueness for positive degrees divisible by 3, excluding degree 6, and cites Machacek. These results do not count the maximum sets in the two nondivisible residue classes. Weighted local covers and their finite-state formulation already appear in [3, Section 3]. Here a particular exact cover is used to control equality and classify all maximum sets; the general covering method is not claimed as new.

Define the proper three-coloring

$$\chi(v) = v_1 - v_2 \pmod 3, \qquad C_j = \{v \in V_d : \chi(v) = j\} \quad (j \in \mathbb{Z}/3\mathbb{Z}),$$

and the body and corners

$$B_d = \{v \in V_d : \max_i v_i \leq d-4\}, \qquad K_i = \{v \in V_d : v_i \geq d-3\} \quad (1 \leq i \leq 3). \tag{2}$$

For $d \geq 10$, the K_i are disjoint and their union is $V_d \setminus B_d$.

Theorem 1.1. *Let $d \geq 10$ and $3 \nmid d$. Every maximum independent set of $G_3(d)$ agrees on B_d with exactly one color class C_j. For each j, all its maximum extensions are obtained by making, independently at the three corners, the choices in Table 1, in the local coordinates of Section 4 with relative color $\delta_i = j - \chi(de_i) \pmod 3$. There are exactly nine extensions for each j, and hence*

$$a_3(d) = 27.$$

Thus Theorem 1.1, together with the cited divisible case, gives (1). The classification is part of the counting argument, rather than an additional assertion requiring a separate proof.

2 An exact triangle cover

Throughout Sections 2–4, assume $d \geq 10$ and $3 \nmid d$, and put $N = |V_d| = \binom{d+2}{2}$. Every edge changes χ, so each C_j is independent. The cyclic permutation $(v_1, v_2, v_3) \mapsto (v_2, v_3, v_1)$ changes χ to $\chi - d$ modulo 3. Consequently the three classes have equal size $N/3$.

Let e_1, e_2, e_3 be the standard basis vectors. For nonnegative integer triples b, define the triangles

$$\begin{aligned} U(b) &= \{b + e_1, b + e_2, b + e_3\}, && b_1 + b_2 + b_3 = d - 1, \\ D(b) &= \{b + e_1 + e_2, b + e_1 + e_3, b + e_2 + e_3\}, && b_1 + b_2 + b_3 = d - 2. \end{aligned}$$

They are three-vertex cliques. Let $\mathcal{T}_d$ be the family of these triangles. Define

$$\rho(t) = \begin{cases} t, & 0 \leq t \leq 3, \\ 4 + ((t-4) \bmod 3), & t \geq 4. \end{cases}$$

The key of $T = U(b)$ or $D(b)$ is

$$(d \bmod 3, \mathrm{type}(T), \mathrm{sort}(\rho(b_1), \rho(b_2), \rho(b_3))).$$

Table 2 gives an integer $W(T)$ for each of the 89 keys that occur, and we set $w(T) = W(T)/60$.

Lemma 2.1 (Exact cover)**.** *Every triangle in $\mathcal{T}_d$ has a key in Table 2, all its weights are nonnegative, and*

$$\sum_{\substack{T \in \mathcal{T}_d \\ v \in T}} W(T) = 60 \qquad (v \in V_d). \tag{3}$$

Proof. The triangles incident with v are exactly

$$U(v - e_i) \quad (v_i > 0), \qquad D(v - e_i - e_j) \quad (i < j,\ v_i, v_j > 0).$$

For $10 \leq d \leq 21$, $3 \nmid d$, direct integer addition using Table 2 verifies all 1134 vertex equations. All incident keys are present, all 89 table entries occur in this range, and their weights are nonnegative. This finite check is implemented in `verify_howroyd_certificate.py`; Section 5 specifies the computations.

If $d > 21$, some coordinate v_i is at least 8. Set $v' = v - 3e_i \in V_{d-3}$. The i-th base coordinate of any incident triangle is either v_i or $v_i - 1$. Before and after subtraction of 3, that coordinate is at least 4, so its ρ-value is unchanged. The existence conditions for all incident triangles are

also unchanged, as are the other coordinates and d mod 3. Thus the incident triangles correspond term by term, with identical keys. Repeated reduction reaches the checked range $10 \le d \le 21$ without dropping below 10. This proves both (3) and the absence of missing keys at larger degrees. □

Corollary 2.2 (Saturation)**.** *The independence number is $N/3$. If I is a maximum independent set, then*

$$|I \cap T| = 1 \qquad \text{for every } T \in \mathcal{T}_d \text{ with } w(T) > 0.$$

Proof. For every independent set I, the exact cover gives

$$|I| = \sum_{T \in \mathcal{T}_d} w(T)|I \cap T| \le \sum_{T \in \mathcal{T}_d} w(T) = \frac{1}{3} \sum_{v \in V_d} \sum_{T \ni v} w(T) = \frac{N}{3}.$$

The color classes attain this bound. For a maximum set,

$$0 = \frac{N}{3} - |I| = \sum_{T \in \mathcal{T}_d} w(T)(1 - |I \cap T|).$$

Every summand is nonnegative, so each triangle with positive weight is met exactly once. □

3 Propagation and constancy on the body

The zero entries of Table 2 have a simple description. If the actual base coordinates are sorted as $s \le t \le u$, then

$$\begin{aligned} W(U(b)) = 0 &\iff (s,t) \in \{(0,1),(1,1),(1,2)\}, \quad u \ge 4, \\ W(D(b)) = 0 &\iff (s,t) \in \{(0,0),(0,2),(1,1)\}, \quad u \ge 4. \end{aligned} \tag{4}$$

This equivalence is checked on the 89 keys. It applies to actual coordinates because $\rho(t) \le 2$ if and only if $t \le 2$, and $\rho(t) \ge 4$ if and only if $t \ge 4$.

Fix a maximum independent set I, and write $f(v) = \mathbf{1}_{\{v \in I\}}$. If two positive-weight triangles share an edge, their saturation equations show that f has the same value at their two opposite vertices. In particular, for $v_i \ge 2$, put

$$F_i(v) = v + (1,1,1) - 3e_i.$$

The triangles $U(v - e_i)$ and $D(v - 2e_i)$ share the vertices $v - e_i + e_j$ for $j \ne i$; their opposite vertices are v and $F_i(v)$. Therefore

$$f(v) = f(F_i(v)) \tag{5}$$

whenever both weights are positive. The move F_i preserves χ.

Lemma 3.1 (Balancing at a maximum coordinate)**.** *Let $v \in B_d$, and choose any i with $v_i = M = \max_j v_j$. Both $U(v - e_i)$ and $D(v - 2e_i)$ have positive weight, and $F_i(v) \in B_d$.*

Proof. We have $M \geq 4$, while the two other coordinates have sum at least 4, since $M \leq d - 4$. The upper triangle has base coordinates $M - 1, v_j, v_k$. Because $M - 1 \geq 3$, the two small coordinates in a zero pattern in (4) would have to be v_j, v_k. Their sum is at least 4, whereas the upper zero patterns have small-coordinate sum at most 3. Thus this weight is positive.

The lower triangle has base coordinates $M-2, v_j, v_k$. If $M \geq 5$, the same argument applies. If $M = 4$, the changed coordinate is 2. A zero pattern not using this 2 is excluded by $v_j + v_k \geq 4$. The only lower zero pattern that can use it is $(0, 2)$; but the third original coordinate is at most 4, which would give $d \leq 4 + 0 + 4 = 8$, a contradiction.

The decreased coordinate of $F_i(v)$ is $M - 2 \leq d - 6$. For either other coordinate v_j, maximality of M gives $2v_j \leq M + v_j \leq d$, hence

$$v_j + 1 \leq \lfloor d/2 \rfloor + 1 \leq d - 4 \qquad (d \geq 10).$$

All new coordinates are nonnegative. Thus $F_i(v)$ remains in B_d. □

Lemma 3.2. *For each $j \in \mathbb{Z}/3\mathbb{Z}$, f is constant on $B_d \cap C_j$.*

Proof. Write $d = 3k + r$, with $r \in \{1, 2\}$; here $k \geq 3$. Whenever the maximum coordinate is $M \geq k + 2$, perform a move from Lemma 3.1. For the nonnegative integer potential $\Phi(v) = v_1^2 + v_2^2 + v_3^2$, the change is

$$\Phi(F_i(v)) - \Phi(v) = 2d - 6M + 6 \leq 2r - 6 < 0.$$

The process therefore terminates, preserving the body, color, and value of f.

At termination all coordinates are at most $k + 1$. For $r = 2$, the only possibilities are the three permutations of $(k, k+1, k+1)$; they have distinct colors. For $r = 1$, the possibilities are the permutations of $(k + 1, k, k)$ and $(k - 1, k + 1, k + 1)$. There is one point of each color in each of these two families. Matching points are joined by a maximum-coordinate move, as in

$$(k + 1, k, k) \longmapsto (k - 1, k + 1, k + 1).$$

These moves are allowed by Lemma 3.1; no decrease of the potential is needed for this final connection. Consequently all starting points of a given color have the same value of f. □

Corollary 3.3 (One color on the body)**.** *There is exactly one $j \in \mathbb{Z}/3\mathbb{Z}$ such that*

$$I \cap B_d = C_j \cap B_d.$$

Proof. For $r = 1$, the triangle $U(k, k, k)$ lies in B_d and has positive weight. For $r = 2$, use $U(k + 1, k, k)$. Indeed $k \geq 3$, their base coordinates exclude the zero patterns (4), and their vertex coordinates are at most $k+2 \leq d-4$. Each triangle has one vertex of each color. The three binary constants from Lemma 3.2 sum to 1 by saturation. Exactly one is therefore 1, and all three color classes occur in the body. □

4 The corner configurations and the count

The graph distance from v to de_i is $d - v_i$. Thus each K_i is a radius-three corner with ten vertices. For vertices in distinct corners their graph distance is at least $d-6 \geq 4$, so there are no edges between corners. Use the following coordinate maps:

$$\begin{aligned} \psi_1(p, q) &= (d - p - q, p, q), \\ \psi_2(p, q) &= (q, d - p - q, p), \\ \psi_3(p, q) &= (p, q, d - p - q). \end{aligned} \tag{6}$$

In each case $p, q \geq 0$ and $p+q \leq 3$ describe the corner; adjacency is given by the six differences $\pm(1, 0), \pm(0, 1), \pm(1, -1)$. The only body vertices adjacent to the corner are on $p + q = 4$. That whole layer lies in B_d: its coordinates are $d - 4, p, q$ with $p, q \leq 4 \leq d - 4$.

Put $c_i = \chi(de_i)$, so $(c_1, c_2, c_3) = (d, -d, 0)$ modulo 3. Direct substitution in (6) gives

$$\chi(\psi_i(p, q)) = c_i + p - q \pmod 3.$$

For a fixed body color j, the selected vertices of the outer layer therefore satisfy

$$p - q \equiv \delta_i \pmod 3, \qquad \delta_i = j - c_i.$$

In particular, relative color 0 means that the background color contains the corner vertex.

Lemma 4.1 (Corner enumeration)**.** *Fix $\delta \in \mathbb{Z}/3\mathbb{Z}$ and the vertices of $p+q = 4$ with $p - q \equiv \delta \pmod 3$. Among independent sets in $p, q \geq 0,\ p + q \leq 3$ having no neighbor among those fixed vertices, all sets of maximum size are exactly those in Table 1.*

Proof. This is a finite enumeration on ten vertices. For $\delta = 0, 1, 2$, respectively, deleting vertices adjacent to the fixed outer layer leaves $8, 7, 7$ available vertices. Enumerate all subsets of the available vertices, discard those containing one of the six adjacency differences, and retain those of largest size. The resulting sets are precisely the seven rows of Table 1. The original certificate verifier implements this enumeration directly from the corner graph and its fixed outer layer. An independent implementation also enumerates all 2^{10} subsets before applying the same independence and compatibility conditions. □

Relative color δ	Maximum size	All maximum corner sets
0	4	$\{(0,0),(0,3),(1,1),(3,0)\}$
1	3	$\{(0,0),(0,2),(2,0)\}$ $\{(0,0),(0,2),(2,1)\}$ $\{(0,2),(1,0),(2,1)\}$
2	3	$\{(0,0),(0,2),(2,0)\}$ $\{(0,0),(1,2),(2,0)\}$ $\{(0,1),(1,2),(2,0)\}$

Table 1: Complete corner choices compatible with the fixed outer layer. Coordinates are (p, q) as in (6).

Proof of Theorem 1.1. Corollary 3.3 determines a unique body color j. Since $3 \nmid d$, the three corner colors $d, -d, 0$ are distinct, and hence the relative colors δ_i are $0, 1, 2$ in some order. The background color has $4, 3, 3$ vertices in these respective corners, so it has $N/3 - 10$ vertices in the body.

Every maximum independent set has size $N/3$. By Lemma 4.1, the sum of its three corner sizes is at most $4 + 3 + 3 = 10$; equality in the total size forces each corner to attain its local maximum. There are consequently $1 \cdot 3 \cdot 3 = 9$ choices for the fixed body color. Conversely, every such choice is compatible with the body; distinct corners have no edges between them. It is an independent set of size $N/3$, hence is maximum. Different choices within one background give different sets, and different backgrounds are already distinct on the body. There are therefore $3 \cdot 9 = 27$ maximum independent sets, with exactly the stated classification. □

5 Computation and reproducibility

The final certificate consists solely of the 89 nonnegative integer numerators in Table 2, also supplied as `howroyd_cover_weights.csv`; every denominator is 60. Floating-point linear programming was used during the search for weights. Neither its output tolerances nor a numerical optimality claim are assumptions of the proof. The original verifier uses integer arithmetic and finite-set operations, reads the CSV, and does not import the search code.

With the two Python programs and CSV in the same directory, run the following commands using Python 3.10 or later and its standard library:

```
python verify_howroyd_certificate.py
python referee_checks.py structural --csv howroyd_cover_weights.csv
python referee_checks.py dp --max-degree 20
```

The structural command uses its default maximum degree 60. The explicit `-csv` argument avoids dependence on the independent checker's original `submission/` directory layout.

The original verifier checks the 89 entries, all 1134 base vertex equations, the exact zero-weight patterns, and the corner enumeration. Its further diagnostics check 34 800 vertex equations at degrees $31, 32, 49, 50, 100, 101, 202$ and 24 684 body moves at nondivisible degrees from 10 through 60. These additional diagnostics are not needed for the all-degree reduction in Lemma 2.1.

The separately written program `referee_checks.py` does not import the original verifier. Its structural mode reconstructs adjacency and finds triangles as graph cliques. For $10 \le d \le 60$, $3 \nmid d$, it checks 25 704 vertex equations, 47 821 triangles, and 25 296 maximum-coordinate moves, including every choice when maxima tie. It also checks the body color components and the corner configurations; for the 16 nondivisible degrees from 10 through 32, it constructs and checks the 27 distinct maximum-size extensions. The structural mode still uses the same weight certificate.

Its full-graph dynamic program is a separate cross-check. It processes rows of lengths $1, 2, \ldots, d+1$, recording the largest size and its number of realizations for each independent row mask. A short-row mask s and the following long-row mask t are compatible exactly when

$$s \,\&\, \bigl(t \mid (t \gg 1)\bigr) = 0,$$

where the operations are bitwise. Each vertex set has a unique row representation. This computation does not use the weights, body classification,

or a prescribed answer 27. It was run for every $d = 0, \ldots, 20$; in particular it gives

$$a_3(7) = 28, \quad a_3(8) = 32, \quad a_3(9) = a_3(12) = a_3(15) = a_3(18) = 1$$

and $a_3(d) = 27$ for $d \in \{10, 11, 13, 14, 16, 17, 19, 20\}$. The original verifier, structural checks, and dynamic program were rerun locally and reproduced the recorded results. These are finite checks. The assertions for every allowed degree use the coordinate reduction and balancing proofs above.

6 Provenance and verification status

AI systems contributed to the mathematical argument, certificate, verification programs, and manuscript preparation. The argument underwent AI-assisted review, and the verification programs were rerun locally with the results described above. The 27-count theorem for all $d \geq 10$ with $3 \nmid d$ has also been formalized in Lean 4.19.0 using mathlib v4.19.0. The source, fixed dependency revisions, and build instructions are provided in the repository's `lean/` directory. The final theorem depends only on Lean's standard axioms `propext`, `Classical.choice`, and `Quot.sound`; it uses no unproved placeholders, custom axioms, or native evaluation. The cited divisible-by-three branch is not part of this formalization. No human expert review or checking by a separate external proof kernel has been completed. The literature search completed for this draft did not locate an earlier proof of the full 27-set statement; that search was not exhaustive and does not establish priority. The references below identify the versions examined. In particular, Machacek's published journal text was not obtained for comparison with the cited arXiv version.

A A boundary-deletion correction in the divisible case

This note concerns only the cited proof of the known divisible case, not Theorem 1.1. On page 5 of the arXiv version of [2], the proof of Theorem 2.1 takes $d = 3k$, identifies a forced boundary independent set J of size $3k = d$, and writes $G''' = G_3(d) \setminus J \cong G_3(d-3)$. With ordinary vertex deletion, this is not literally correct. The entire boundary

$$\partial V_d = \{v \in V_d : \min_i v_i = 0\}$$

has $3d = 9k$ vertices for $d > 0$. Deleting only J therefore leaves $6k$ more vertices than $G_3(d-3)$.

The required correction is local to that step. The preceding argument on each boundary side X_i gives $|I \cap X_i| = k+1$ and forces $J \cap X_i \subseteq I \cap X_i$ with $|J \cap X_i| = k+1$. Hence

$$I \cap \partial V_d = J.$$

Delete the whole boundary and put

$$H = G_3(d)[V_d \setminus \partial V_d].$$

Subtracting $(1,1,1)$ gives $H \cong G_3(d-3)$. Equation (2.3) of [2] then gives

$$|I \setminus J| = \alpha_3(d) - 3k = \alpha_3(d-3) = \alpha(H).$$

Thus $I \setminus J$ is a maximum independent set of H, and the stated induction continues: its inductive step has $k > 3$, so $d - 3 \geq 9$. This is an explicit correction to the wording of the examined version, not a quotation of a correction made by its author or a new attribution of the uniqueness theorem.

B The complete integer certificate

In Table 2, $r = d \bmod 3$, τ is the triangle type, and $a \leq b \leq c$ are the sorted ρ-values of its base coordinates, not the unmodified coordinates. The final column is $W = 60w$. The table contains all 89 rows of `howroyd_cover_weights.csv`.

Table 2: Integer numerators of the exact triangle cover.

r	τ	a	b	c	W
1	D	0	0	5	0
1	D	0	1	4	30
1	D	0	2	6	0
1	D	0	3	5	5
1	D	0	4	4	10
1	D	0	5	6	3
1	D	1	1	6	0
1	D	1	2	5	22
1	D	1	3	4	13

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Table 2 (continued)

r	τ	a	b	c	W
1	D	1	4	6	15
1	D	1	5	5	16
1	D	2	2	4	3
1	D	2	3	3	24
1	D	2	3	6	17
1	D	2	4	5	12
1	D	2	6	6	10
1	D	3	3	5	3
1	D	3	4	4	24
1	D	3	5	6	10
1	D	4	4	6	17
1	D	4	5	5	17
1	D	5	6	6	17
1	U	0	0	6	60
1	U	0	1	5	0
1	U	0	2	4	30
1	U	0	3	6	30
1	U	0	4	5	25
1	U	0	6	6	32
1	U	1	1	4	0
1	U	1	2	6	0
1	U	1	3	5	3
1	U	1	4	4	4
1	U	1	5	6	3
1	U	2	2	5	16
1	U	2	3	4	3
1	U	2	4	6	8
1	U	2	5	5	6
1	U	3	3	3	3
1	U	3	3	6	17
1	U	3	4	5	3
1	U	3	6	6	17
1	U	4	4	4	3
1	U	4	5	6	3
1	U	5	5	5	3
1	U	6	6	6	3
2	D	0	0	6	0
2	D	0	1	5	30
2	D	0	2	4	0
2	D	0	3	6	3
2	D	0	4	5	3

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Table 2 (continued)

r	τ	a	b	c	W
2	D	0	6	6	6
2	D	1	1	4	0
2	D	1	2	6	24
2	D	1	3	5	15
2	D	1	4	4	6
2	D	1	5	6	8
2	D	2	2	5	3
2	D	2	3	4	24
2	D	2	4	6	17
2	D	2	5	5	3
2	D	3	3	3	3
2	D	3	3	6	3
2	D	3	4	5	10
2	D	3	6	6	3
2	D	4	4	4	3
2	D	4	5	6	3
2	D	5	5	5	3
2	D	6	6	6	3
2	U	0	0	4	60
2	U	0	1	6	0
2	U	0	2	5	30
2	U	0	3	4	30
2	U	0	4	6	27
2	U	0	5	5	30
2	U	1	1	5	0
2	U	1	2	4	0
2	U	1	3	6	3
2	U	1	4	5	9
2	U	1	6	6	7
2	U	2	2	6	12
2	U	2	3	5	3
2	U	2	4	4	3
2	U	2	5	6	17
2	U	3	3	4	3
2	U	3	4	6	10
2	U	3	5	5	3
2	U	4	4	5	17
2	U	4	6	6	17
2	U	5	5	6	17

References

[1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A297557. Howroyd's conjecture dated 1 January 2018 and Machacek's comment dated 1 May 2021. `https://oeis.org/A297557/internal`. Consulted 13 October 2026.

[2] John Machacek, *Unique maximum independent sets in graphs on monomials of a fixed degree*, arXiv:2010.11112v2, 14 March 2021. `https://arxiv.org/abs/2010.11112v2`.

[3] Aryeh Lev Zabokritskiy (Yohananov), *Independent Sets in Multiset Profile Graphs via Weighted Local Covers*, arXiv:2607.13733v3, 4 August 2026. `https://arxiv.org/abs/2607.13733v3`.

[4] Avraham Kreindel, Isaac Barouch Essayag, and Aryeh Lev Zabokritskiy (Yohananov), *Multiset Deletion Codes: Cyclic Constructions, Bounds, and Exact Results*, arXiv:2601.05636v2, 13 July 2026. `https://arxiv.org/abs/2601.05636v2`.